

Lab 4: Arithmetic Logic Unit

The purpose of the lab is to build an Arithmetic Logic Unit (ALU). An ALU is a combinational circuit that can perform mathematical operations and is an essential building block for most digital devices (https://en.wikipedia.org/wiki/Arithmetic_logic_unit).

- 1) Design an ALU. It should have **eight** functions: **addition**, **subtraction**, and six others of your choice including at least **one bitwise** and **one shift** operation (Check the list on the Wikipedia page).
- 2) You are expected to use “block design” functionality in Vivado. You must implement the ALU using **small reusable modules**, then build larger blocks from them.
 - a. For example, you can write VHDL code for a “full adder” and using only that code block multiple times you can create an 8-bit adder.
 - b. Build steps:
 - i. Write your VHDL code.
 - ii. Press “Create Block Design” under “IP integrator”.
 - iii. Right click an empty space, click “Add Module”.
 - iv. You can copy-paste modules or write multiple VHDL files.
 - v. Connect modules appropriately.
 - vi. Use “Create Port” or “Make External” to connect your design to switches and LEDs.
 - vii. Right click on your block design under “Sources” tab and press “Create HDL Wrapper”.
 - viii. Write simulation testbench and simulate your design.
 - ix. Write constraint file and generate bitstream.
- 3) Put the **simulation results** and **block design** in your report.
- 4) Deploy your ALU on your **FPGA** and display the outputs through LEDs. Show your schematics and working FPGA to your TA to get their approval. Include representative photos of your FPGA in your report.

Do not use operators from libraries. Implement your operations yourself.